# Chaofan Lin

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## Education

**Tsinghua University**, Beijing, China — Sep. 2024 – Present

*Ph.D. Candidate in Computer Science.* Advisor: Mingyu Gao

- A member of Institute for Interdisciplinary Information Sciences (IIIS).

**Shanghai Jiao Tong University**, Shanghai, China — Sep. 2020 – Jun. 2024

*Bachelor in Computer Science.* Advisor: Yong Yu, Weinan Zhang

- A member of **ACM Honors Class**, an elite CS program in SJTU.
- **GPA:** 94 / 100 | **Ranking: 1** / 35.
- **Selected Courses:** Compiler 100/100, Operating System 100/100, Machine Learning 97/100, Mathematical Logic 100/100, Advanced Compiler 100/100, Algorithm 98/100. (And other 20 A+ courses)

My research interest lies in building practical, scalable and efficient systems for machine learning (MLSys), including algorithm-system co-design (Twilight **[1]**), serving system (ParrotServe **[2]**) , and deep learning compiler (Core maintainer @ TileLang, contributors @ apache/tvm).

## Selected Publications

**[1] Twilight: Adaptive Attention Sparsity with Hierarchical Top-$p$ Pruning.**

**Chaofan Lin**, Jiaming Tang, Shuo Yang, Hanshuo Wang, Tian Tang, Boyu Tian, Ion Stoica, Song Han, Mingyu Gao. *In the Thirty-Ninth Annual Conference on Neural Information Processing Systems* **(NeurIPS 25 Spotlight)**
[arXiv] | [Code]

**[2] Parrot: Efficient Serving of LLM-based Applications with Semantic Variable.**

**Chaofan Lin**, Zhenhua Han, Chengruidong Zhang, Yuqing Yang, Fan Yang, Chen Chen, Lili Qiu. *In 18th USENIX Symposium on Operating Systems Design and Implementation* **(OSDI 24)**
[Hugging Face Daily Papers] | [arXiv] | [Code] | [conference page]

**[3] Accelerating Large-Scale Reasoning Model Inference with Sparse Self-Speculative Decoding.**

Yilong Zhao*, Jiaming Tang*, Kan Zhu, Zihao Ye, Chi-Chih Chang, **Chaofan Lin**, Jongseok Park, Guangxuan Xiao, Mohamed S. Abdelfattah, Mingyu Gao, Baris Kasikci, Song Han, Ion Stoica. *In the Thirty-Ninth Annual Conference on Neural Information Processing Systems* **(MLSys 26)**
[arXiv] | [Code]

## Research Experience

**DeepSeek AI** *Research Intern* — 2025.11 - Present

I'm a core maintainer of TileLang. My main scopes include:

- Its multi-backend support part. We will design a unified plugin interface to support multiple backends.
- Its IR (Intermediate Representation) design and implementation, with tight collaboration with the TVM community. We will embrace TIR-X, which is the next generation tensor IR in TVM.
- Host part of the compiler, including mitigating the overhead of host overhead through Host CodeGen.
- Tutorials (TileLang-Puzzles), documentation and user-friendly supports.

**Seed Foundation Team, Bytedance** *R&D Intern* — 2024.9 - 2025.10

- Developed a unified kernel library for sparse attention algorithms, including Twilight **[1]**.
- Involved in the development of a multi-hardware kernel compiler TritonX.

**System and Networking Group, Microsoft Research Asia** *Research Intern* 2023.6 - 2024.6

- Lead the project Parrot **[2]**, which is a serving system for LLM applications / LLM-agents. Parrot proposes Semantic Variable, a unified abstraction to expose application-level knowledge to public LLM services, hence opens a brand new optimization space for LLM inference serving.

**Catalyst, Carnegie Mellon University** *Research Intern, Remote* 2022.7 - 2023.5

Advised by Tianqi Chen. Topic: Deep Learning Compiler.

- Developed a training workflow for Relax (the next-gen graph-level IR of TVM), including registration mechanism of operator gradients and an automatic differentiation pass. (20+ PRs to apache/tvm)
- Helped to deploy a simple LLM on device through machine learning compilation (MLC). And also contributed to a project which aims to perform fine-tuning (LoRA) on device.

## Talks

**Cross Platform Training Using Automatic Differentiation on Relax IR** 2023

At TVM Conference 2023. [Video Record]

## Honors and Awards

**Scholarships**

- Comprehensive Excellence Scholarship of Tsinghua University. (*First-level, the highest*) 2025
- Zhiyuan Outstanding Scholarship. (*Top 30 winnners in Zhiyuan Honor College*) 2024
- OSDI Student Travel Grant. 2024
- **National Scholarship.** (*Top 0.2% national-wide.*) 2022, 2023
- Foresight-Sequoia Talent Development Fund. (*5 winners each year in SJTU.*) 2021
- Zhiyuan Undergraduate Excellence Award. *A-level, the highest.* 2021, 2022
- Zhiyuan Honorary Scholarship. 2020, 2021, 2022

**Honors**

- **Best Bachelor Thesis in SJTU.** *Top 1%* 2024
- Outstanding Graduates in SJTU. 2024

**Competitions**

- The Chinese Mathematics Competitions (Shanghai Region). *First Prize.* 2022
- Mathematical Contest In Modeling and Interdisciplinary Contest In Modeling. *Meritorious Winner.* 2021

## Side Projects

**Triton-Puzzles-Lite** 700+ stars, 2024

Puzzles for learning Triton, playable with minimal environment setup. I'm building this because many Triton beginners can't find good introductory materials.

**ACMClass Online Judge** 2022

Previously maintained ACMClass OJ, an online judge for SJTU students, providing efficient and stable OJ services for courses like the ACM and IEEE classes.

## Teaching

**Teaching Assistant**

- **Introduction to Computer System, Tsinghua University**, *Spring 2025, With Prof. Wei Xu and Prof. Mingyu Gao*: I lead the TA team to help distribute Raspberry Pi devices and give talks to the Yao Class preparatory course.
- **Advanced Compiler, Shanghai Jiao Tong University**, *Spring 2023, With Prof. Yong Yu*: I gave several talks on Polyhedra model and Register Allocation in this course. [Lecture Notes]
- **Mathematical Logic, Shanghai Jiao Tong University**, *Fall 2022, With Prof. Qiang Yin and Yijia Chen*
- **Programming Design (A), Shanghai Jiao Tong University**, *Fall, 2021, With Prof. Huiyu Weng*: I help students in this course implement a simple Python 3 Interpreter.

## Skills

- Programming Languages: Python, C/C++, Java, Verilog, Go, Web (HTML, CSS, JavaScript), LaTeX.
- Kernel Programming/Deep Learning Compilers: CUDA C, Triton, TVM.
- English: CET-6 600, CET-4 661.

Last Updated: Jun. 2026